O’ZBEKISTON RESPUBLIKASI

OLIY TA’LIM FAN VA INNOVATSIYALAR VAZIRLIGI

TOSHKENT AXBOROT TEXNOLOGIYALARI UNIVERSITETI

DASTURIY INJINERING FAKULTETI
“ALGORITMLASH VA MATEMATIK MODELLASHTIRISH’’ KAFEDRASI

“Matematik dasturlash ” fanidan

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1. Sistemaning nomanfiy bazis yechimini toping

$$\begin{cases} x_1 + 2x_2 - x_3 - x_4 = 1, \\ -x_1 + 2x_2 + 3x_3 + x_4 = 2, \\ x_1 + 5x_2 + x_3 - x_4 = 5, \end{cases}$$

$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0.$$

2. Masalani basis yechimlaridan birini toping va uni optimal yechimga aylantiring

$$\begin{cases} -2x_1 + x_2 + x_3 = 2, \\ x_1 - 2x_2 + x_4 = 2, \\ x_1 + x_2 + x_5 = 5, \end{cases}$$

$$x_j \geq 0, \ (j = 1,2,3,4,5),$$

$$Y = -x_1 + x_2 \to \min$$

3. Quyidagi ochiq modelli transport masalasini yeching

b_i / a_i	3	3	3	2	2
4	3	2	1	2	3
5	5	4	3	1	1
7	0	2	3	4	5

4. Berilgan butun sonli dasturlash masalasini grafik usulda yeching

$$\begin{cases} x_1 + x_2 \leq 6, \\ 3x_1 + 10x_2 \leq 26, \end{cases}$$

$$x_1 \geq 0, x_2 \geq 0,$$

$$x_1, x_2 - butun.$$

Skaszej kopsbirg

① Sis tomamy nomanfiy bazis yechimini, dgm

$$\begin{cases} x_1 + 2x_2 - x_3 - x_4 = 1 \\ -x_1 + 2x_2 + 3x_3 + x_4 = 2 \\ x_1 + 5x_2 + x_3 - x_4 = 5 \\ x_1 \geq 0 \; x_2 \geq 0 \; x_3 \geq 0 \; x_4 \geq 0 \end{cases}$$

1) $\boxed{x_4 = 0}$

$$\begin{cases} x_1 + 2x_2 - x_3 = 1 \\ -x_1 + 2x_2 + 3x_3 = 2 \\ x_1 + 5x_2 + x_3 = 5 \end{cases} \Rightarrow \begin{cases} 4x_2 + 2x_3 = 3 & (1) \text{ sa } (2) \text{ ni qo'sh} \\ -3x_2 - 2x_3 = -4 \\ 3x_2 + 2x_3 = 4 & (1) \text{ sa } (3) \text{ ni qo'sh} \end{cases}$$

$$\begin{cases} 4x_2 + 2x_3 = 3 \\ -3x_2 + 2x_3 = 4 \end{cases} \quad x_2 = -1 \quad \boxed{x_3 = 3,5}$$

$x_1 = 1 + 3,5 + 2 \cdot 1$ $\quad x_1 = 6,5$

$x_2 < 0 \Rightarrow$ nomanfiy yechim garagari.

2) $\boxed{x_3 = 0}$

$$\begin{cases} x_1 + 2x_2 - x_4 = 1 \\ -x_1 + 2x_2 + x_4 = 2 \\ x_1 + 5x_2 - x_4 = 5 \end{cases} \quad \begin{matrix} 4x_2 = 3 & x_2 = \frac{3}{4} & (1)+(2) \\ 7x_2 = 7 & & (2)+(3) \end{matrix}$$

$x_2 = 1$ $\quad 3x_2 = 4$ $\quad x_2 = \frac{4}{3}$ $\quad (1)+(3)$

x_2 lar har xil bunda ham yechim yo'q

3) $\boxed{x_1 = 0}$

$$\begin{cases} 2x_2 - x_3 - x_4 = 1 \\ 2x_2 + 3x_3 + x_4 = 2 \\ 5x_2 + x_3 - x_4 = 5 \end{cases} \quad \begin{cases} 4x_2 + 2x_3 = 3 & (1)+(2) \\ 3x_2 + 2x_3 = 4 & (3)-(1) \end{cases}$$

$x_2 = -1 \Rightarrow$ nomanfiy ϕ

4) $x_2 = 0$

$$\begin{cases} x_1 - x_3 - x_4 = 1 \\ -x_1 + 3x_3 + x_4 = 2 \\ x_1 + x_3 - x_4 = 5 \end{cases}$$

$2x_3 = 3$ $\quad x_3 = \frac{3}{2}$ $\quad (1)+(2)$

$4x_3 = 7$ $\quad x_3 = \frac{7}{4}$ $\quad (2)+(3)$

$-2x_3 = -4$ $\quad x_3 = 2$ $\quad (1)-(3)$

nomanfiy bazis yechimi mavjud emas.

② $$\begin{cases} -2x_1 + x_2 + x_3 = 2 \\ x_1 - 2x_2 + x_4 = 2 \\ x_1 + x_2 + x_5 = 5 \\ Y = -x_1 + x_2 \to \min \end{cases}$$

$-2 \cdot x + 1 = 0$ $\quad x = \frac{1}{2}$

$1 \cdot \frac{1}{2} + 2 = -\frac{3}{2}$

$-2 \cdot x + 1 = 0$ $\quad x = \frac{1}{2}$

$-2 \cdot x + 1 = 0$ $\quad x = \frac{1}{2}$

	x_1	x_2	x_3	x_4	x_5	b
x_3	-2	1	1	0	0	2
x_4	1	-2	0	1	0	2
x_5	1	1	0	0	1	5
Y	1	-1	0	0	0	0

	x_1	x_2	x_3	x_4	x_5	b
x_1	1	$-\frac{1}{2}$	$-\frac{1}{2}$	0	0	-1
x_4	0	$-\frac{3}{2}$	$\frac{1}{2}$	1	0	3
x_5	0	$\frac{3}{2}$	$\frac{1}{2}$	0	1	6
Y	0	$-\frac{1}{2}$	$\frac{1}{2}$	0	0	1

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	x_1	x_2	x_3	x_4	x_5	b
x_3	0	-3	1	2	0	6
x_1	1	-2	0	1	0	2
x_5	0	3	0	-1	1	3
Y	0	1	0	-1	0	-2

$1 \cdot x - 2 = 0$ $\quad x = 2$

$$\begin{cases} 0 \cdot -2 + 1 = -3 \\ 0 \cdot 2 + 1 = 1 \\ 1 \cdot 2 + 0 = 2 \\ 0 \cdot 2 + 0 = 0 \\ 2 \cdot 2 + 2 = 6 \end{cases}$$

$$\begin{cases} 1 + x + 1 = 0 \quad x = -1 \\ -2 \cdot -1 + 1 = 3 \\ 0 \cdot -1 + 0 = 0 \\ 1 \cdot -1 + 0 = -1 \\ 0 \cdot -1 + 1 = 1 \end{cases}$$

$$\begin{cases} 1 \cdot x + 1 = 0 \quad x = -1 \\ -2 \cdot -1 - 1 = 1 \\ 0 \cdot -1 + 0 = 0 \\ 1 \cdot -1 + 0 = -1 \\ 0 \cdot -1 + 0 = 0 \end{cases}$$

	x_1	x_2	x_3	x_4	x_5	b
x_3	0	0	1	1	1	3
x_1	1	0	0	$\frac{1}{3}$	$\frac{2}{3}$	4
x_2	0	1	0	$-\frac{1}{3}$	$\frac{1}{3}$	1
Y	0	0	0	$-\frac{2}{3}$	$-\frac{1}{3}$	-3

$3 \cdot x - 3 = 0$ $\quad x = 1$

$3 \cdot x - 2 = 0$ $\quad x = \frac{2}{3}$

$$\begin{cases} 1 \cdot \frac{2}{3} + 1 = \frac{1}{3} \\ 1 \cdot \frac{2}{3} + 0 = \frac{2}{3} \\ 3 \cdot \frac{2}{3} + 2 = 4 \end{cases}$$

$$\begin{cases} 3 \cdot x + 1 = 0 \quad x = -\frac{1}{3} \\ -1 \cdot -\frac{1}{3} - 1 = -\frac{2}{3} \\ 1 \cdot -\frac{1}{3} + 0 = -\frac{1}{3} \\ 1 \cdot -\frac{1}{3} - 2 = -3 \end{cases}$$

$x_1 = 4 \quad x_2 = 1 \quad x_3 = 3$

$x_4 = x_5 = 0 \quad Y_{max} = -3$

$-2 \cdot 4 + 1 + 3 = 2$

(3) $a = (4, 5, 7) \Rightarrow$ Taminot

$b = (3, 3, 3, 2, 2) \Rightarrow$ Talab

$$C = \begin{pmatrix} 3 & 2 & 1 & 2 & 3 \\ 5 & 4 & 3 & 1 & 1 \\ 0 & 2 & 3 & 4 & 5 \end{pmatrix}$$

$\sum a_i = 4 + 5 + 7 = 16$

balans

$\sum b_i = 3 + 3 + 3 + 2 + 2 = 13$

$16 \neq 13$ balansyoq

$16 - 13 = 3$

$a_i \backslash b_i$	3	3	3	2	2	3
4	3	2	1	2	3	0
5	5	4	3	1	1	0
7	0	2	3	4	5	0

	b_1	b_2	b_3	b_4	b_5	b_6
$a_i \backslash b_i$	3	3	3	2	2	3
4	3	1	~~4~~			
5		2	3			
7				2	2	3

$x_{11} = \min(4, 3) = 3$

$$\begin{cases} b_1 = 3 - 3 = 0 \quad a_1 = 4 \quad b_2 = 3 \\ a_1 = 4 - 3 = 1 \end{cases}$$

$$\begin{cases} x_{12} = \min(1, 3) = 1 \\ a_1 = 1 \quad b_2 = 3 \end{cases}$$

$$\begin{cases} a_1 = 1 - 1 = 0 \\ b_2 = 3 - 1 = 2 \end{cases}$$

$x_{22} = \min(5, 2) = 2$

$$\begin{cases} b_2 = 2 - 2 = 0 \\ a_2 = 5 - 2 = 3 \end{cases}$$

$$\begin{cases} x_{23} = \min(3, 3) = 3 \\ b_3 = 3 - 3 = 0 \\ a_2 = 3 - 3 = 0 \end{cases}$$

$$\begin{cases} x_{34} = \min(7, 2) = 2 \\ b_4 = 2 - 2 = 0 \quad a_3 = 7 - 2 = 5 \end{cases}$$

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$$\begin{cases} x_{35} = \min(5,2) = 2 \\ b_5 = 2-2=0 \\ a_3 = 5-2=3 \end{cases}$$

$$\begin{cases} x_{36} = \min(8,3) = 3 \\ b_6 = 3-3=0 \\ a_3 = 3-3=0 \end{cases}$$

	Minimum	Maxx	Zinu
x_{11}	3	3	8
x_{12}	1	2	2
x_{22}	2	4	8
x_{23}	3	3	8
x_{32}	2	4	8
x_{35}	2	5	10
x_{36}	3	0	0

$Z = 3\cdot 2 + 3 + 3\cdot 8 + 10\cdot 0$

$Z = 46$

4

$$\begin{cases} x_1 + x_2 \le 6 \\ 3x_1 + 10x_2 \le 26 \\ x_1 \ge 0 \quad x_2 \ge 0 \\ x_1, x_2 \Rightarrow \text{tutuq son} \end{cases}$$

$$\begin{cases} x_1 = 0 \quad x_2 = 6 \quad x_2 = 0 \quad x_1 = 6 \\ x_1 = 0 \quad x_2 = 2{,}6 \quad x_2 = 0 \quad x_1 = \frac{26}{3} \end{cases}$$

$$\begin{cases} x_1 + x_2 = 6 \quad /\cdot 3 \\ 3x_1 + 10x_2 = 26 \end{cases}$$

$7x_2 = 26 - 18$

$x_2 = \frac{8}{7} \quad x_1 = 6 - \frac{8}{7} = \frac{34}{7} \quad x_2 = 1\frac{1}{7} \quad x_1 = 4\frac{6}{7}$

$(0,6)\ (6,0)\ (0,2{,}6)\ (\frac{26}{3},0)\ (\frac{34}{7},\frac{8}{7})$

$x_1 = 4 \quad x_2 = 1 \quad x_1 + x_2 \le 6 \quad 3x_1 + 10x_2 \le 26$

$3\cdot 4 + 10\cdot 1 = 22 \le 26$

$x_1 = 5 \quad x_2 = 1 \quad 5 + 1 = 6 \quad 3\cdot 5 + 10\cdot 1 = 25$

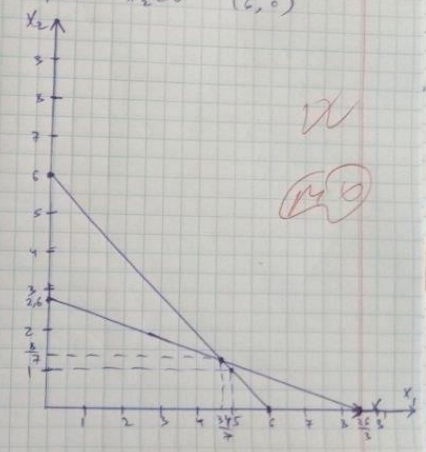
$x_1 = 6 \quad x_2 = 0 \quad 6 + 0 = 6 \quad 3\cdot 6 + 0 = 18$

$x_1 = 0 \quad x_2 = 2 \quad 0 + 2 = 2 \quad 3\cdot 0 + 10\cdot 2 = 20$

$x_1 = 4 \quad x_2 = 2 \quad 4 + 2 = 6 \quad 3\cdot 4 + 10\cdot 2 = 32$

$Z = x_1 + x_2 \Rightarrow max \quad x_1 = 5 \quad x_2 = 1 \quad (5,1)$

$\boxed{Z_{max} = 6}$ $\quad x_1 = 6 \quad x_2 = 0 \quad (6,0)$



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